

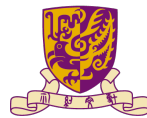
Network Programming

Data Center Networking

Minchen Yu

SDS@CUHK-SZ

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香港中文大學(深圳)
The Chinese University of Hong Kong, Shenzhen



SCHOOL OF
DATA SCIENCE
數據科學學院

Data Center Overview

Cloud datacenter



Copywrite: Google

Data centers

- >100k servers
- Costs in billions of dollars
- Geographically distributed





Image from Microsoft Azure

Data center and network

- Thousands of routers
- Hundreds of thousands of servers

MICROSOFT TECH FACEBOOK

Microsoft and Facebook just laid a 160-terabits-per-second cable 4,100 miles across the Atlantic

47

Enough bandwidth to stream 71 million HD videos at the same time

By Thuy Ong | @ThuyOng | Sep 25, 2017, 7:56am EDT

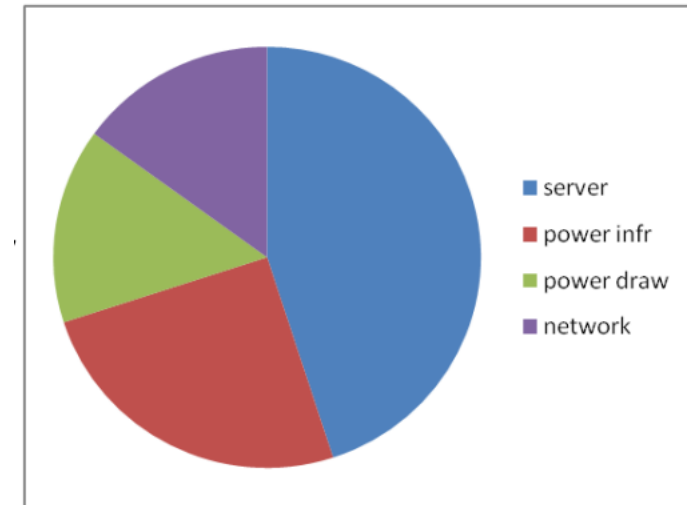
<https://www.nytimes.com/interactive/2019/03/10/technology/internet-cables-oceans.html>

Data center costs

Amortized Cost*	Component	Sub-Components
~45%	Servers	CPU, memory, disk
~25%	Power infrastructure	UPS, cooling, power distribution
~15%	Power draw	Electrical utility costs
~15%	Network	Switches, links, transit

Total cost varies

- Over \$1B for mega data center
- Server costs dominate
- Network costs significant



Server costs

Uneven application fit

- Each server has CPU, memory, disk: most applications exhaust one resource, stranding the others

Long provisioning timescales

- New servers purchased quarterly at best (Uncertainty in demand)
- Demand for a new service can spike quickly (Risk management)
- Not having spare servers to meet demand brings failure

Session state and storage constraints

- If the world were stateless servers, life would be good

30% utilization considered “good” in data centers

Agility: any service, any server

Turn the servers into a single large fungible pool

- Dynamically expand and contract service footprint as needed

Workload management

- Means for rapidly installing a service's code on a server
- Virtual machines, containers, etc.

Storage Management

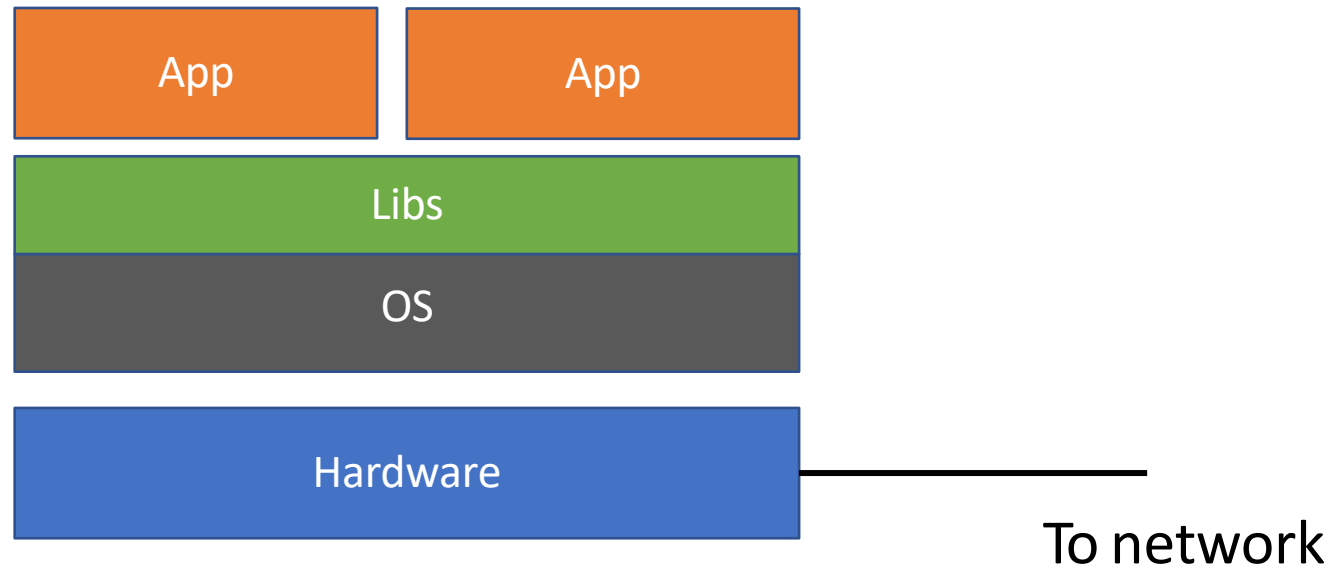
- Means for a server to access persistent data
- Distributed filesystems (e.g., HDFS)

Network

- Means for communicating with other servers, regardless of where they are in the data center

Data Center Servers and Network

Server (originally)



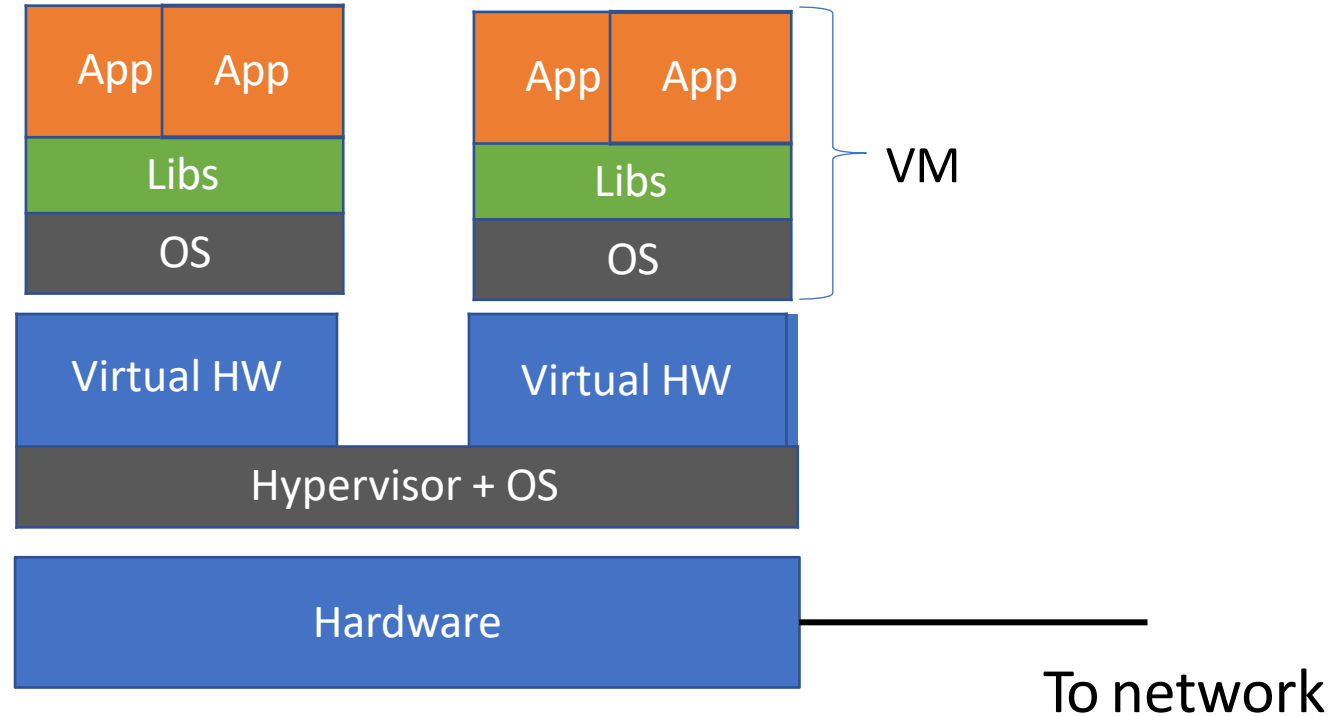
Then came virtual machines (VMs)

HW became too powerful

- Run multiple OSes on the same machine
- Cheaper that way

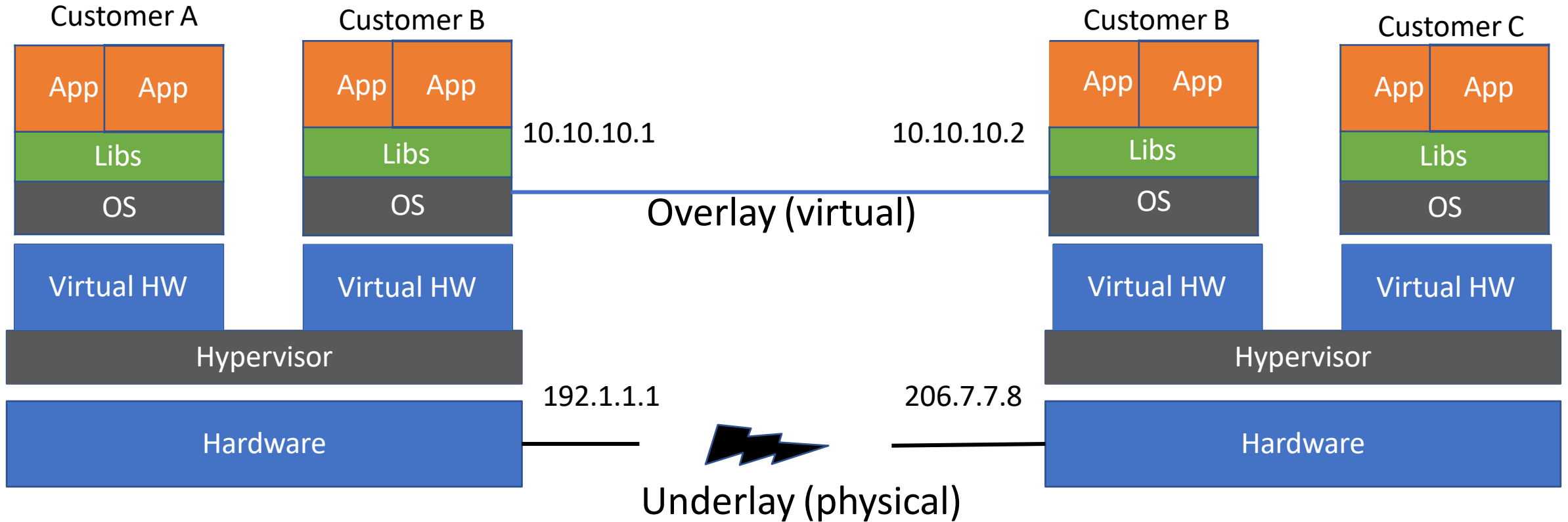
The hypervisor virtualizes the HW and fools the OS

- Provides isolation



The network thinks multiple hosts are connected
The hypervisor acts as a hub for inter-VM traffic

VMs in the cloud



Forwarding between VMs involves a lookup from overlay address to underlay location

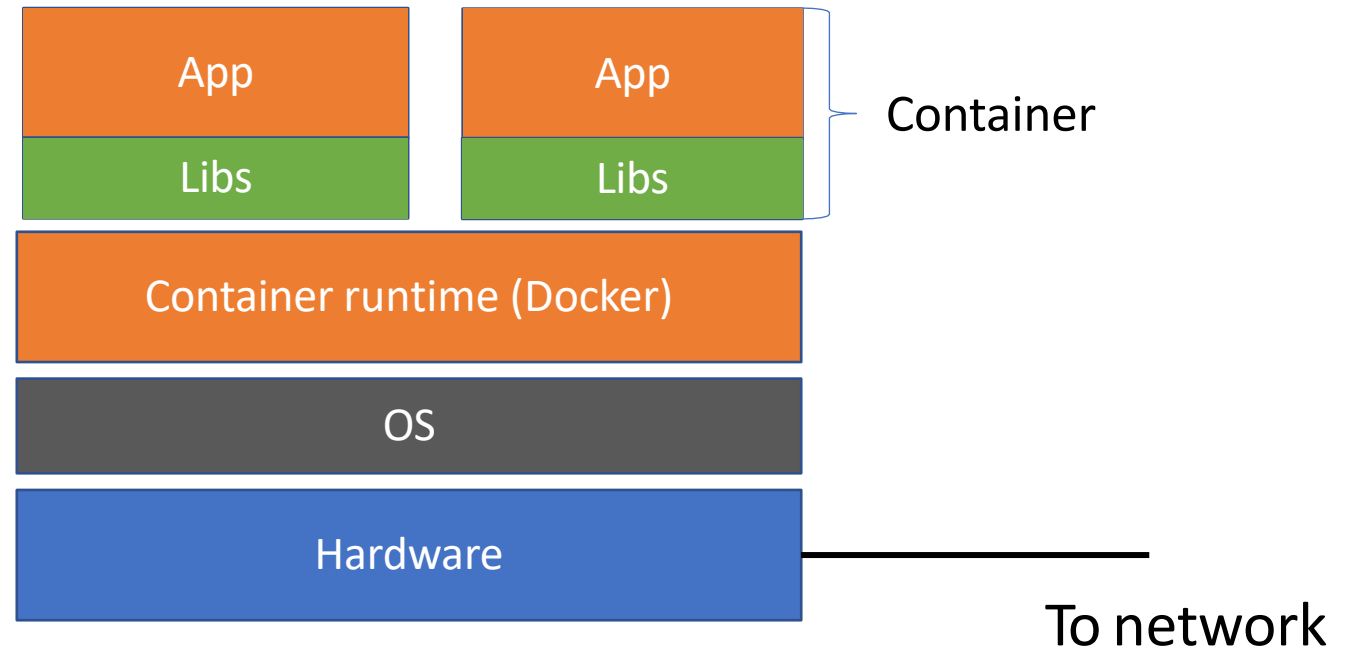
Enter containers

Lighter-weight virtualization than VMs

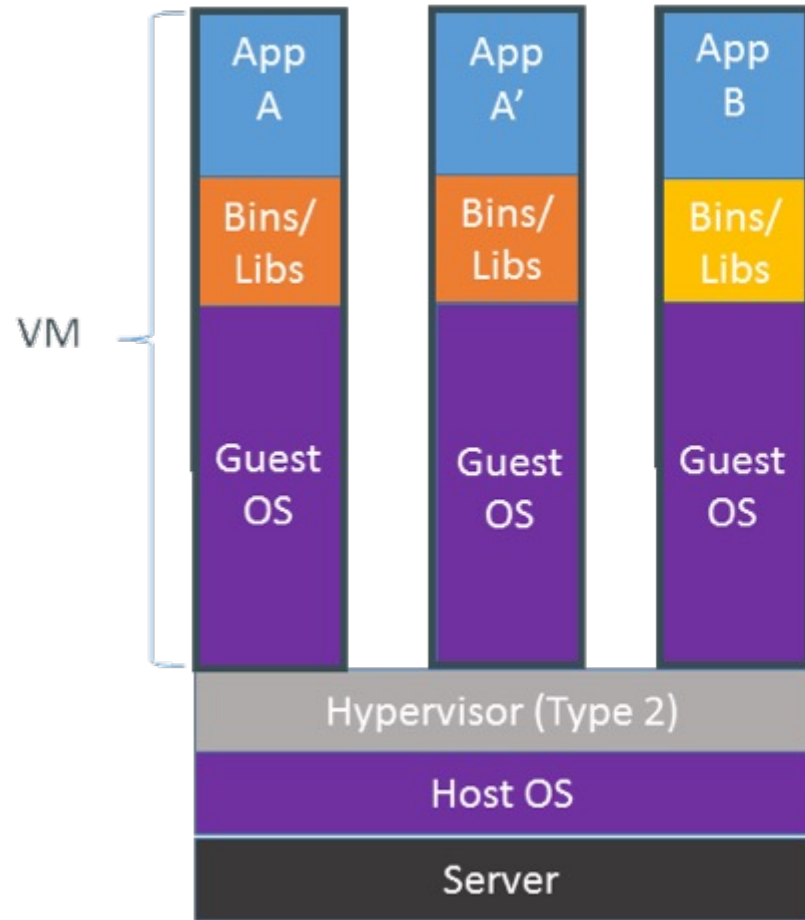
- Libraries, not the full OS

Better isolation and packaging than apps

- Bundle the library versions you need

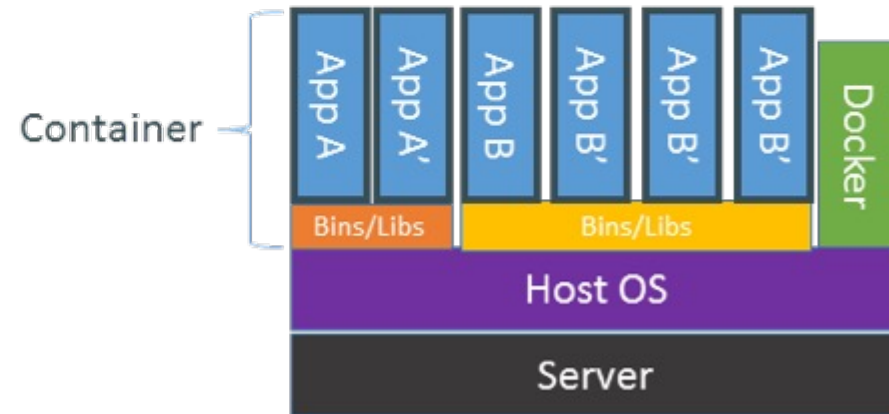


VM vs. Containers

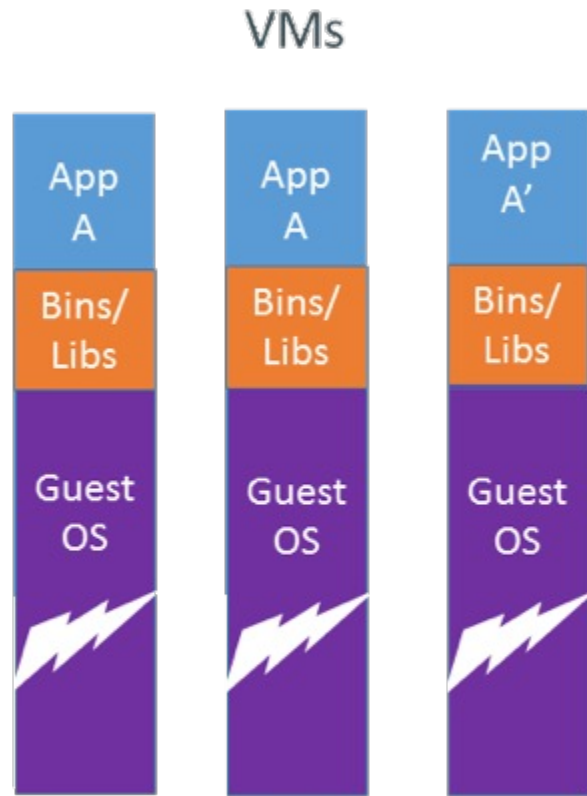


Containers are isolated, but share OS and, where appropriate, bins/libraries

...result is significantly faster deployment, much less overhead, easier migration, faster restart



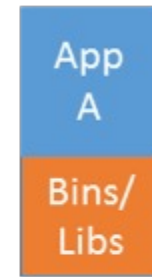
Containers are lightweight



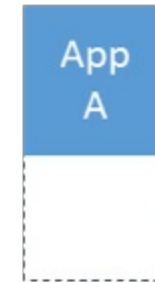
VMs

Every app, every copy of an app, and every slight modification of the app requires a new virtual server

Containers



Original App
(No OS to take up space, resources, or require restart)



Copy of App
No OS. Can Share bins/libs

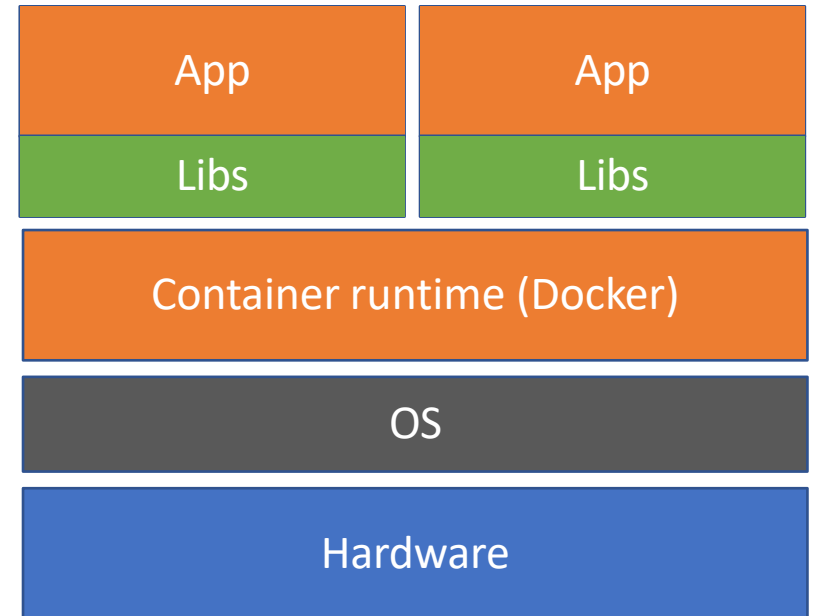


Modified App

Copy on write capabilities allow us to only save the diffs Between container A and container A'

Container networking

- Connect containers to the outside world and to each other
- Port conflicts among containers and other apps running on the same host
- High performance between containers on the same host
- (Virtual) private network between related containers



Container networking: Host

Containers share the IP address (and networking stack) of the host.

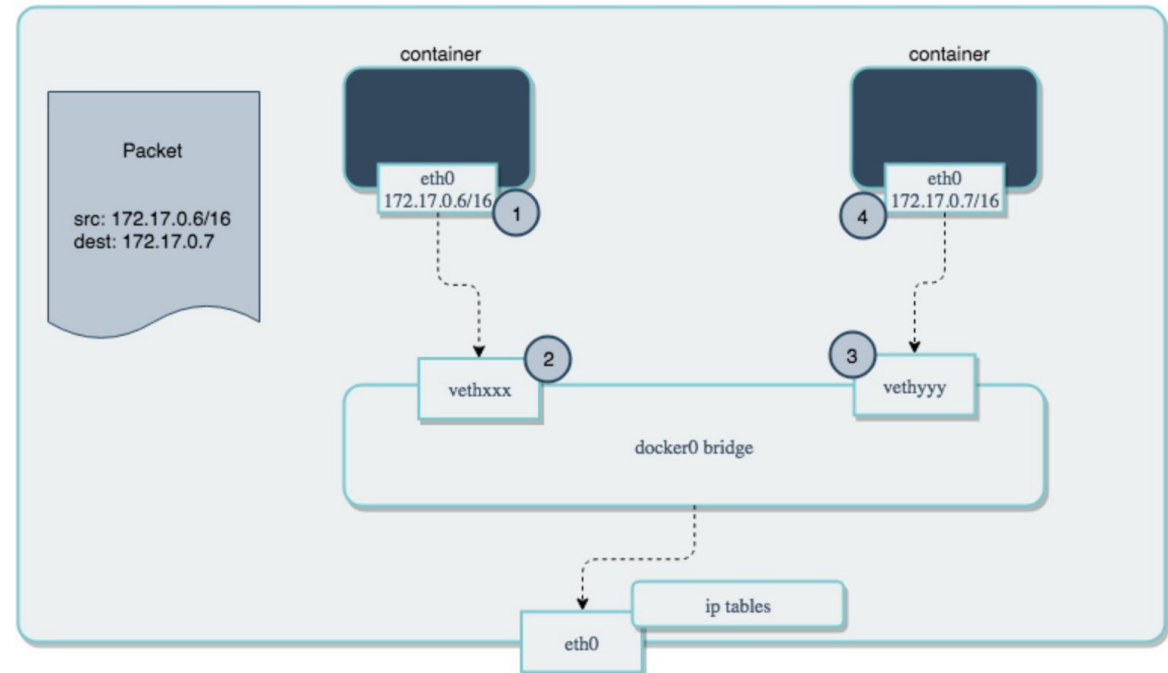
- Cannot handle port conflicts
- Minimal overhead



Container networking: Bridge

An internal network for containers on the same host.

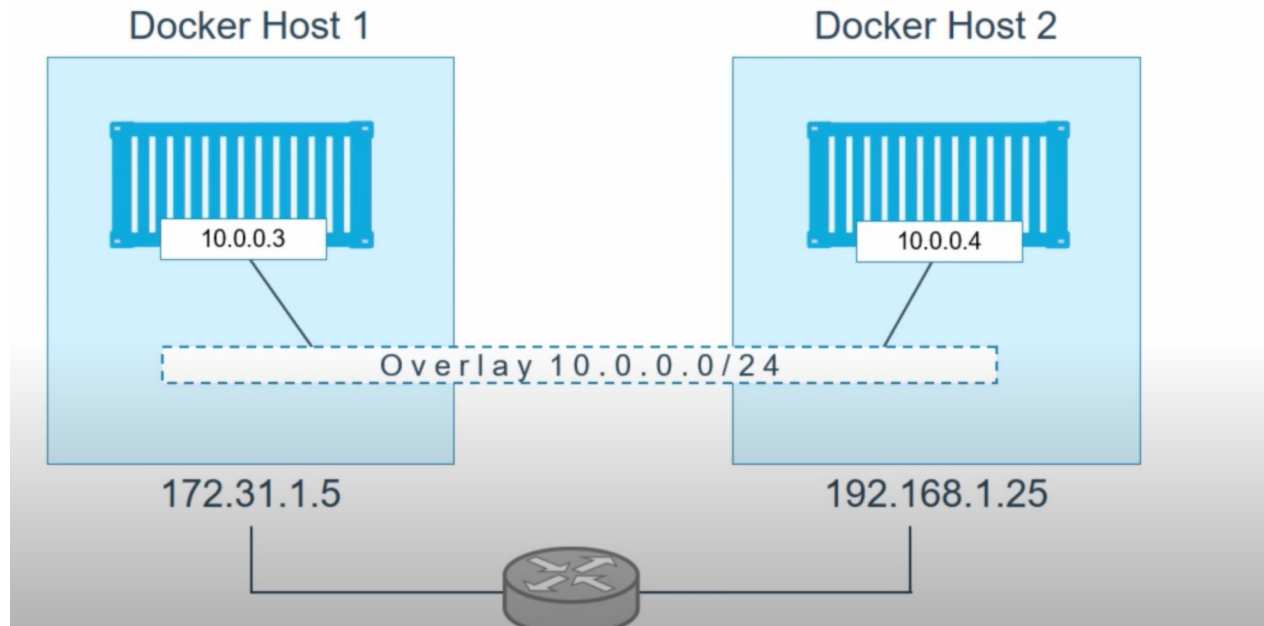
- Use NATs for outside world



Container networking: Overlay

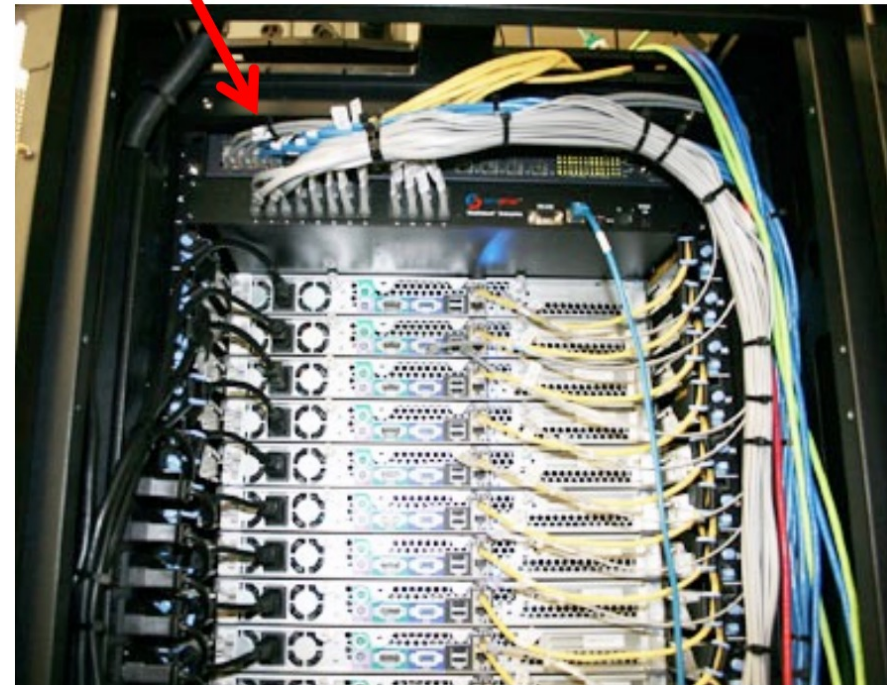
Create a private network across containers on different hosts

- VXLAN is a common way to do that

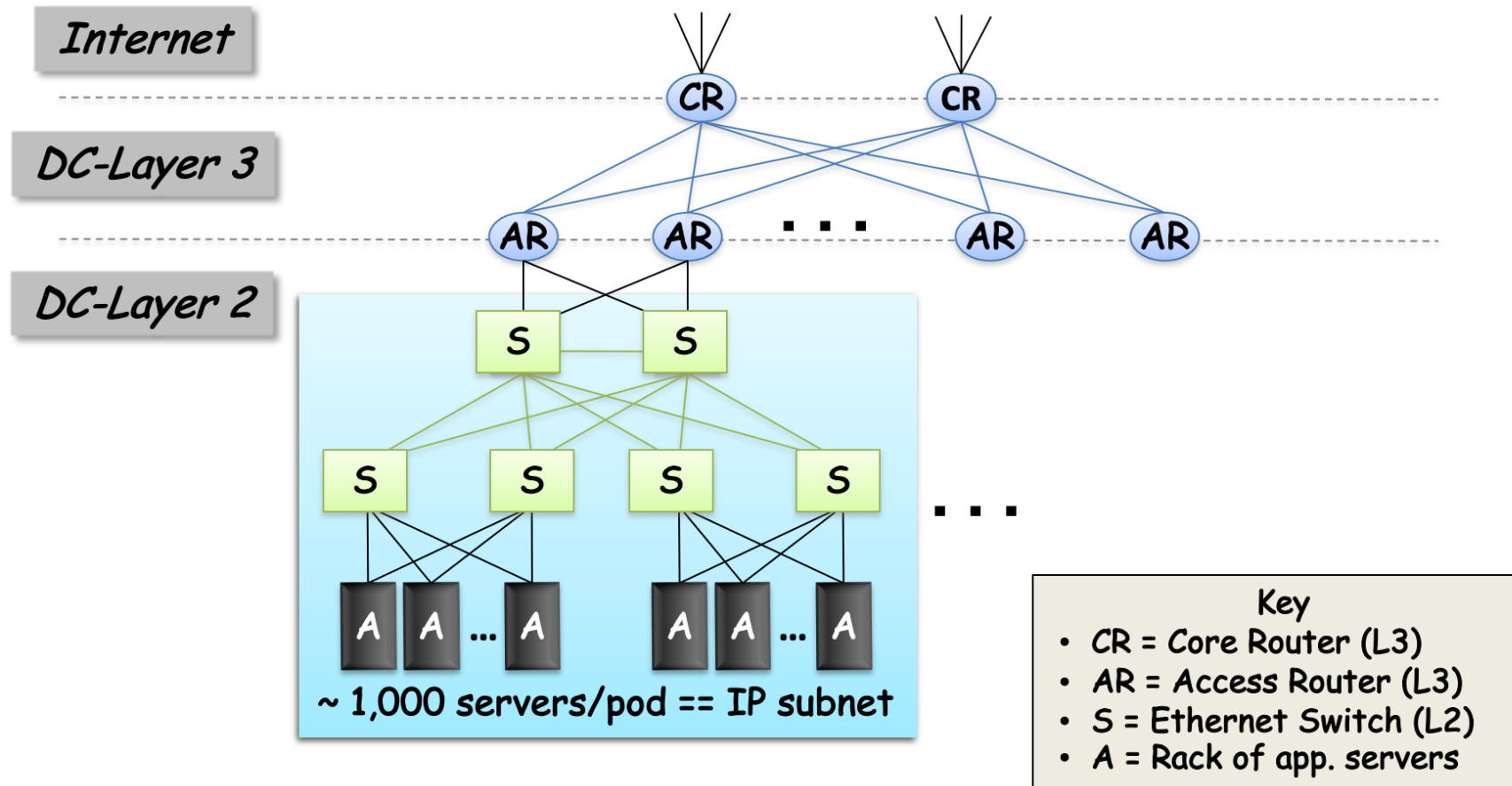


Rack of servers

- Rack of servers
 - Commodity servers, and top-of-rack switch
- Modular design
 - Preconfigured racks
 - Power, network, and storage cabling

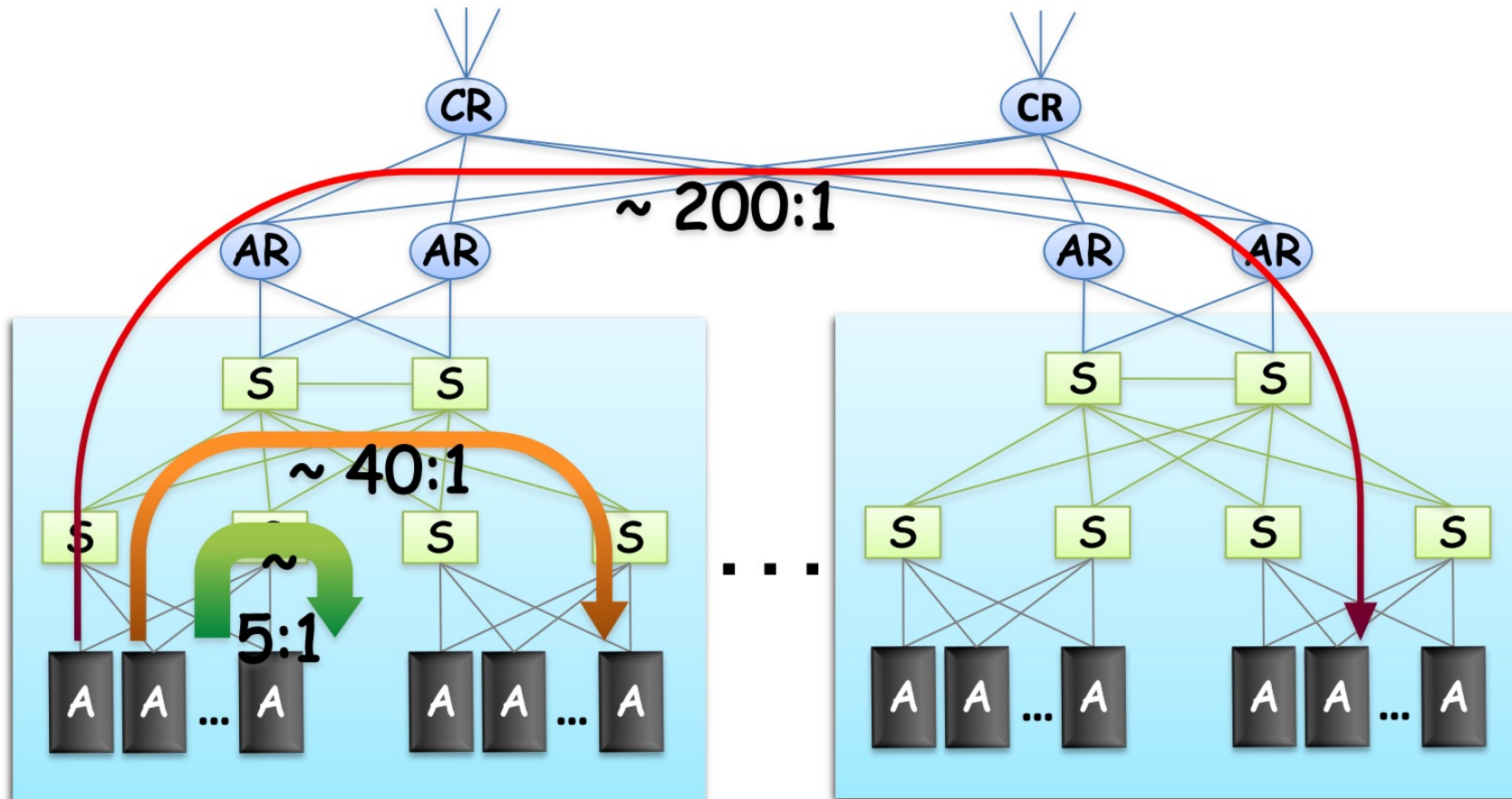


Data center network topology



“Data Center: Load balancing Data Center Services”, Cisco 2004

Oversubscription



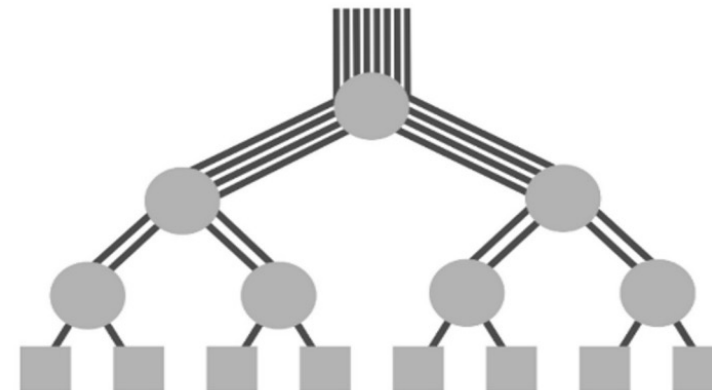
Oversubscription: much more demand vs. supply for higher links

Network bandwidth measurements

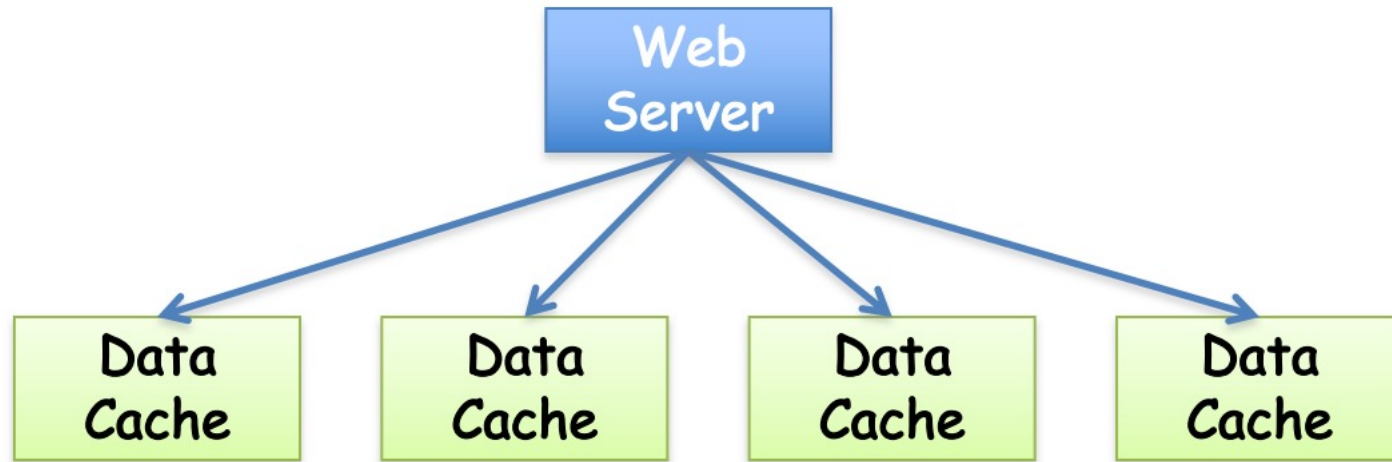
- Bisection bandwidth: Split nodes into two halves such that bandwidth between the halves is minimal, that is the bisection b/w
- Full bisection bandwidth: $\frac{1}{2}$ of the nodes can communicate simultaneously with the other $\frac{1}{2}$ without blocking

Fat trees

- For any switch, the number of links *going down* to its siblings is equal to the number of links *going up* to its parent in the upper level
- If oversubscribed, the number of uplinks is less than the number of downlinks
- ***Non-blocking***: the number of uplinks and downlinks are in proportion of 1:1
 - blocking factor = # downlink/ # uplink

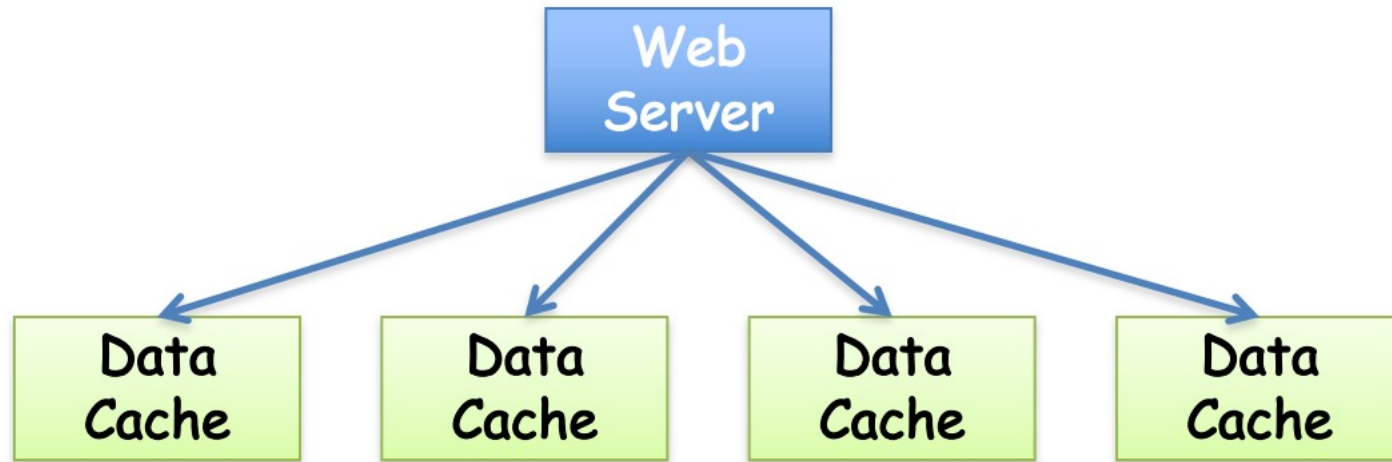


Network Incast



- Incast arises from synchronized parallel requests
 - Web server sends out parallel request (“which friends of Alice are online?”)
 - Nodes reply at same time, cause traffic burst
 - Replies potentially exceed switch’s buffer, causing drops

Network Incast

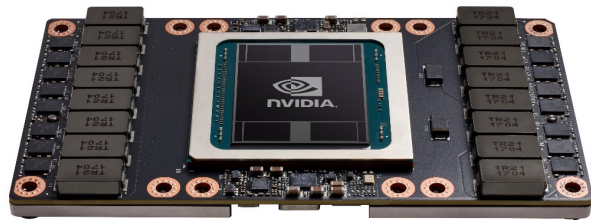


- Solutions mitigating network incast
 - Reduce TCP's min retransmission timeout (often use 200ms >> DC RTT)
 - Increase buffer size
 - Add small randomized delay at node before reply

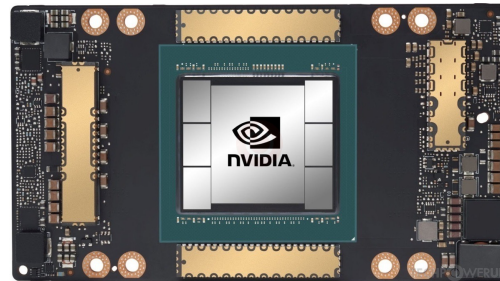
Emerging GPU clusters

GPU

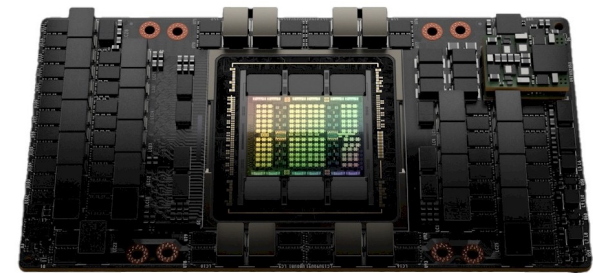
- GPU = Graphics Processing Unit
 - Accelerator for raster based graphics and deep learning
 - Highly programmable
 - Commodity hardware
 - 100s of ALUs; 10s of 1000s of concurrent threads



NVIDIA Volta: V100



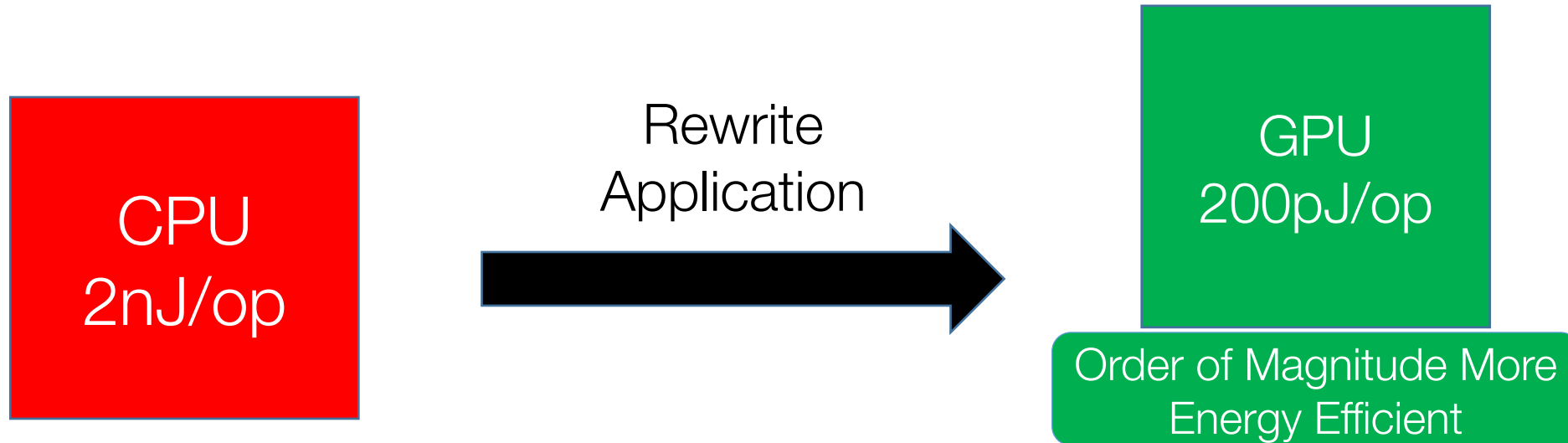
NVIDIA Ampere: A100



NVIDIA Hopper: H100

Why GPU?

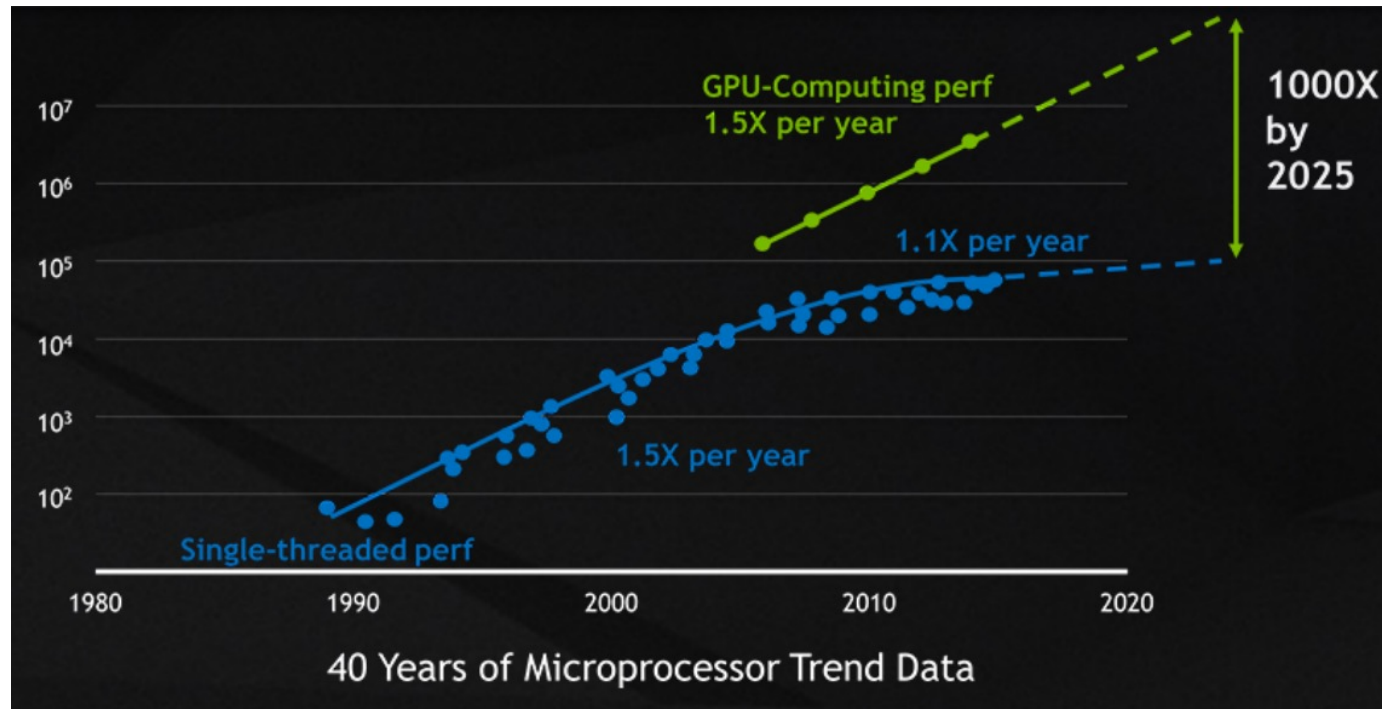
- GPU uses larger fraction of silicon for computation than CPU.
- At peak performance GPU uses order of magnitude less energy per operation than CPU.



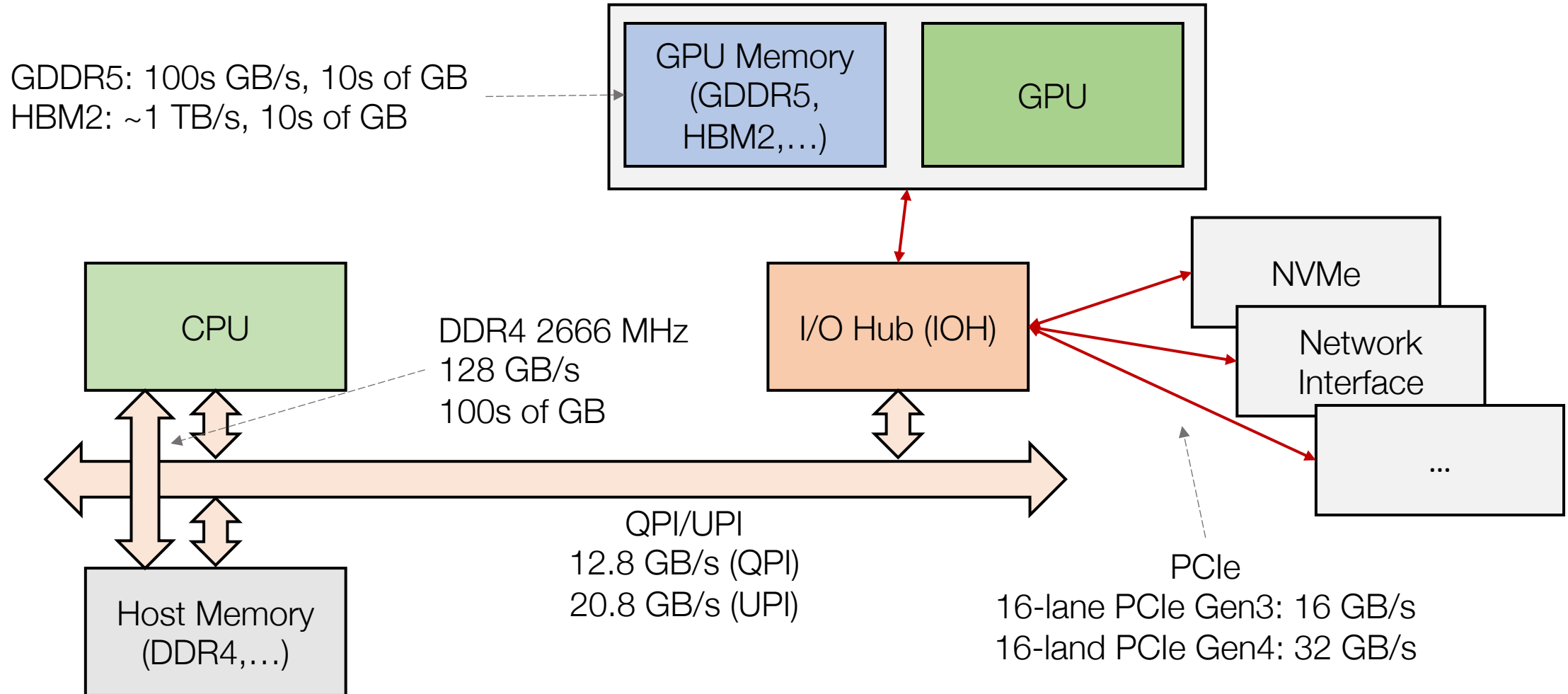
Peak Performance vs. CPU

	Throughput	Power	Throughput/Power
Intel Skylake	128 SP GFLOPS/4 Cores	100+ Watts	~1 GFLOPS/Watt
NVIDIA V100	15 TFLOPS	200+ Watts	~75 GFLOPS/Watt

Also,



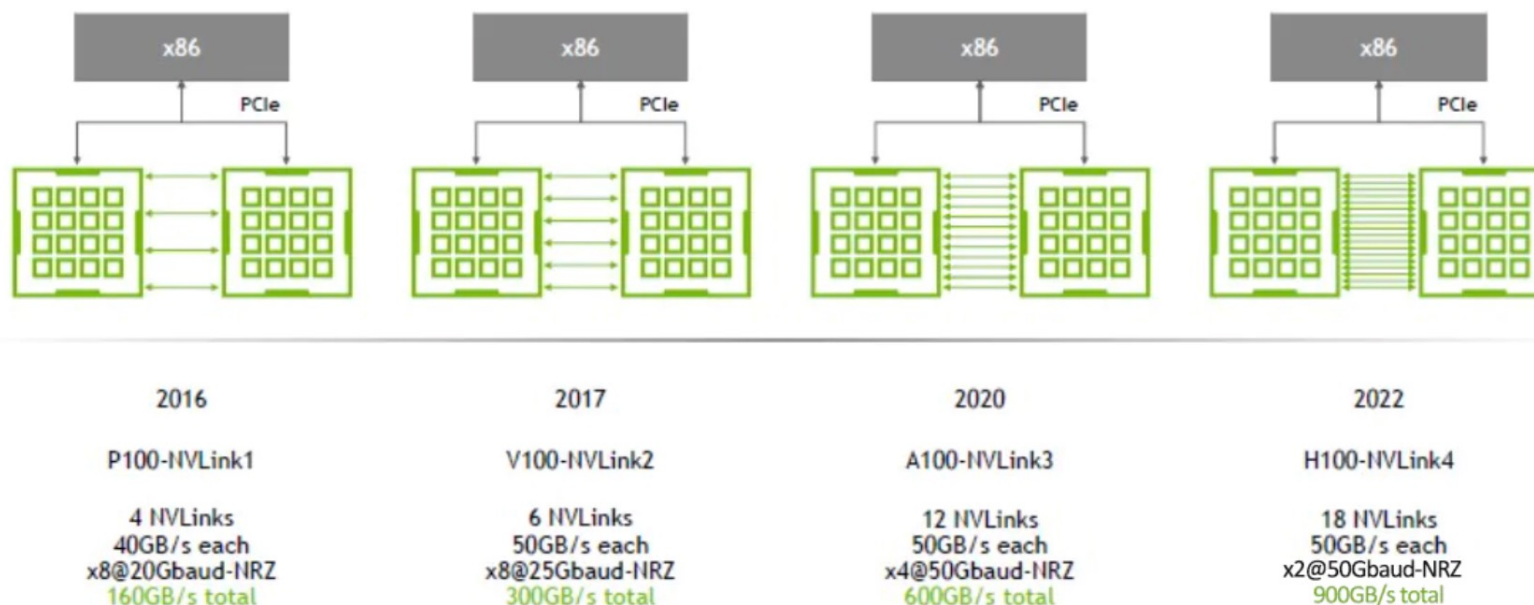
System Architecture Snapshot With a GPU



High-speed interconnect between GPUs

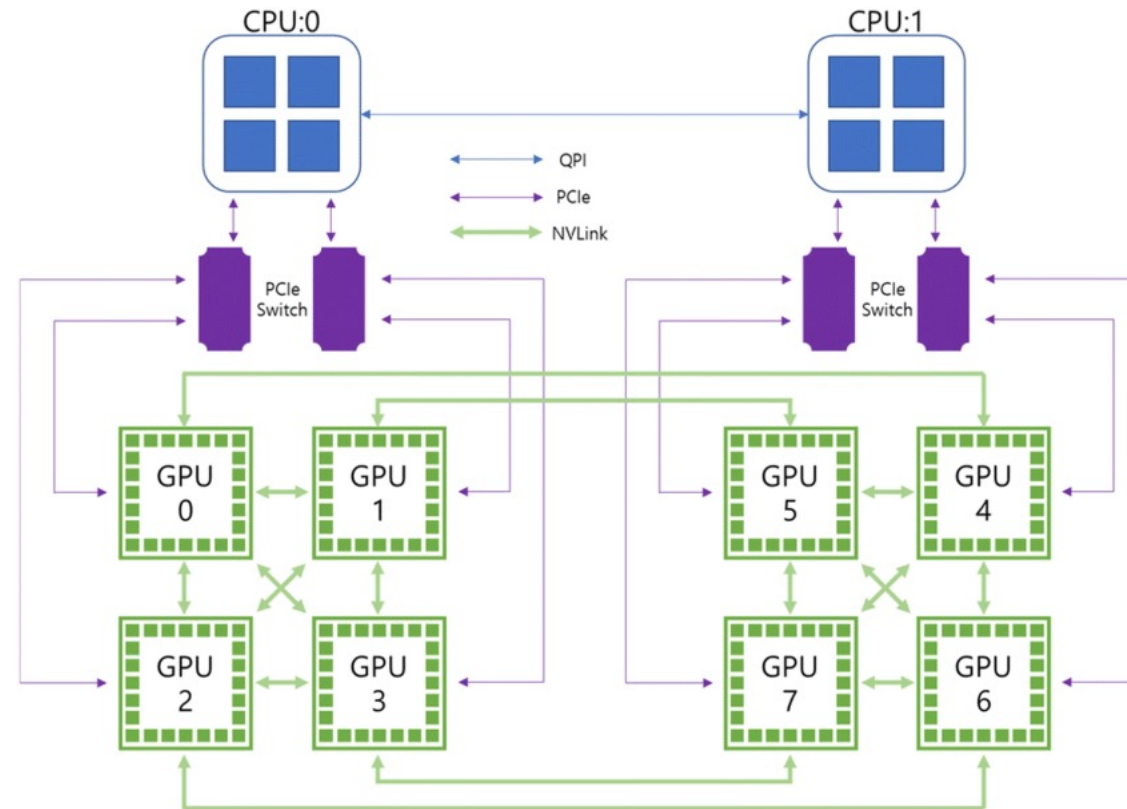
NVLink

- bidirectional, direct GPU-to-GPU interconnect



Multi-GPU topology

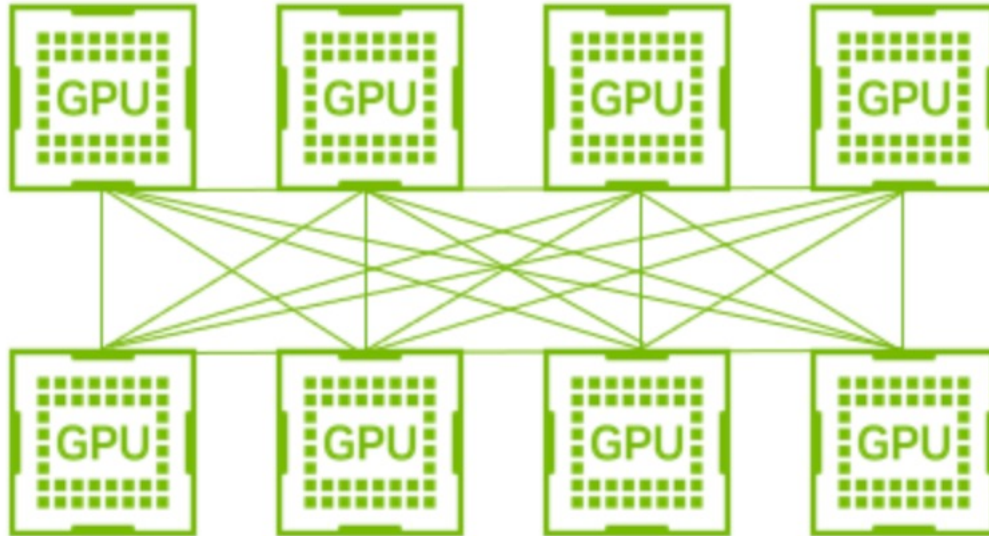
Multi-GPU topology before DGX-A100



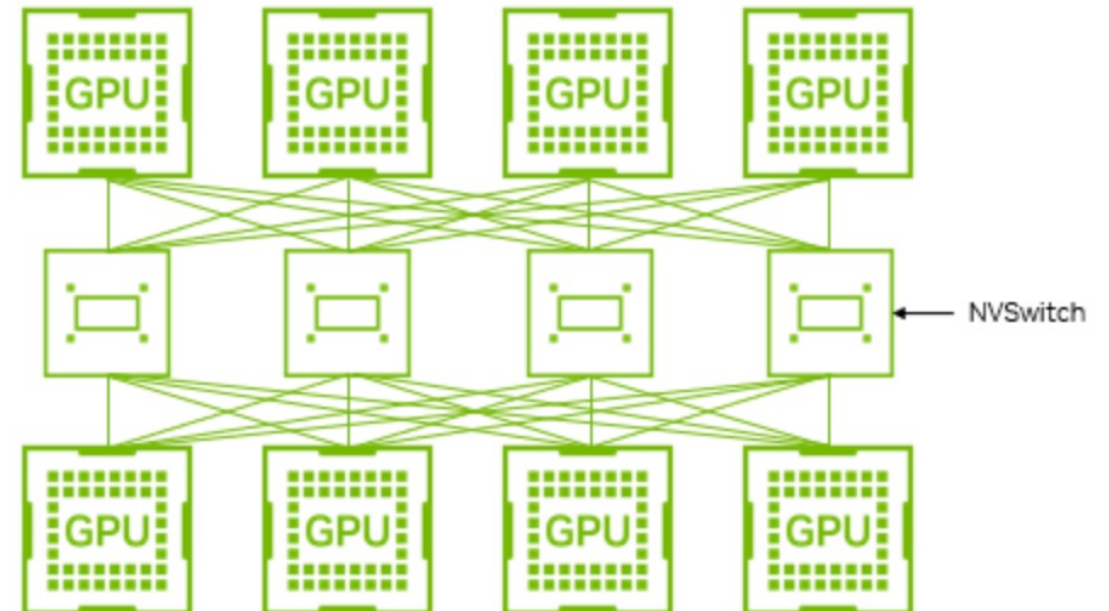
Multi-GPU topology

NVSwitch (NVLink Switch) connects multiple NVLinks to provide all-to-all GPU communication at full NVLink speed

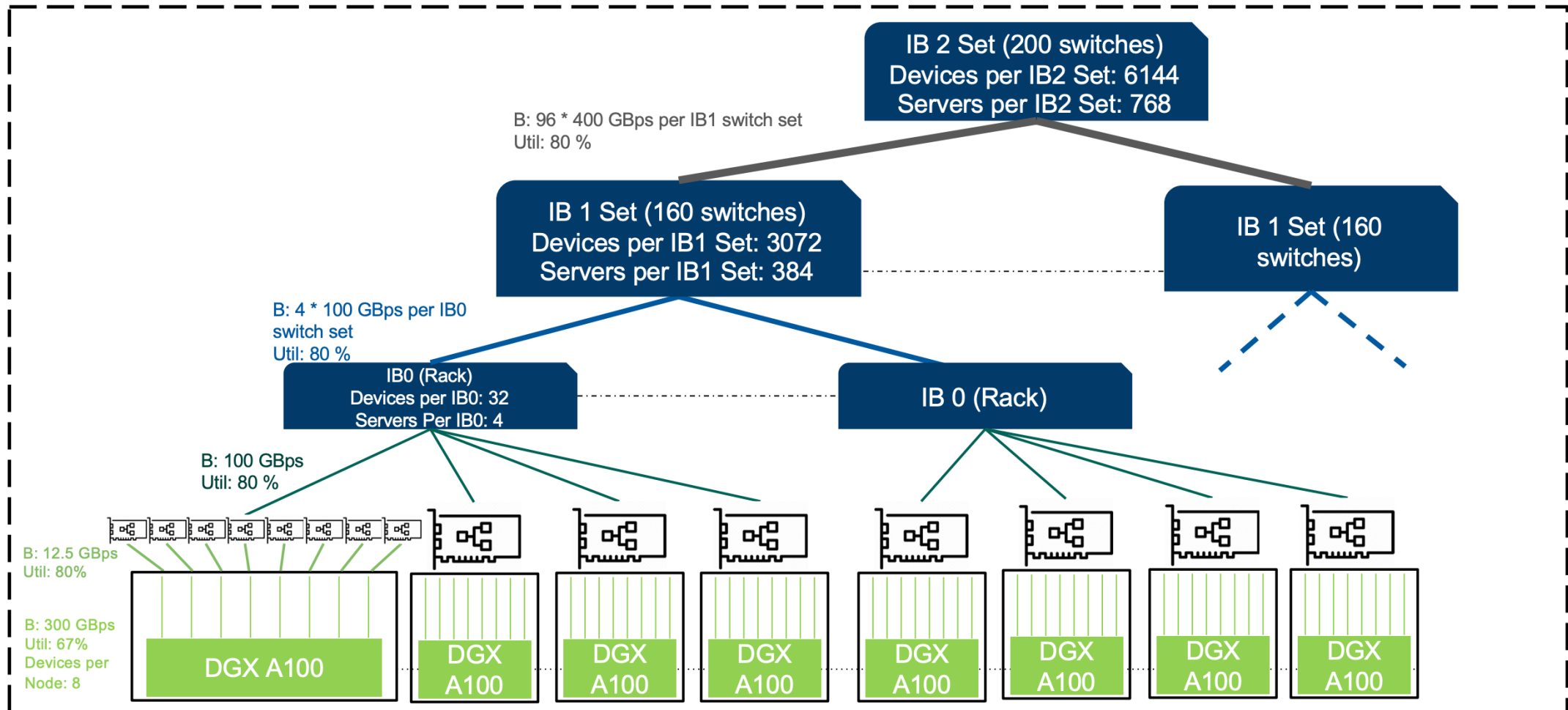
Multi-GPU Configuration without NVSwitch



Multi-GPU Configuration with NVSwitch

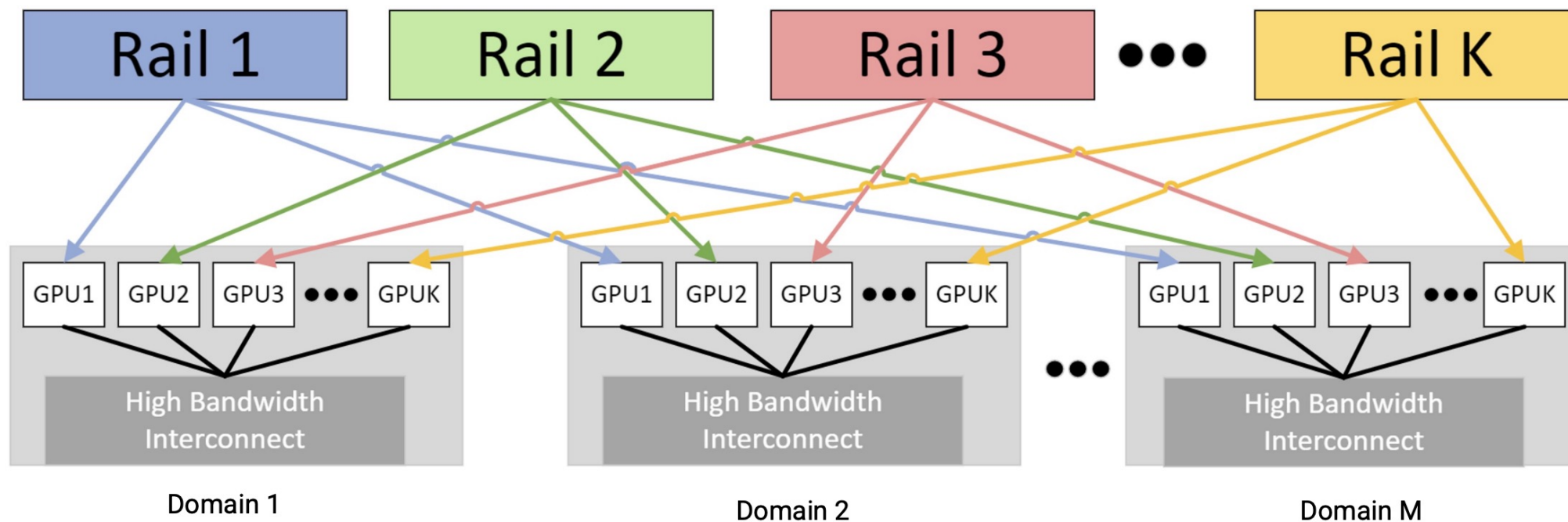


GPU clusters: Fat trees



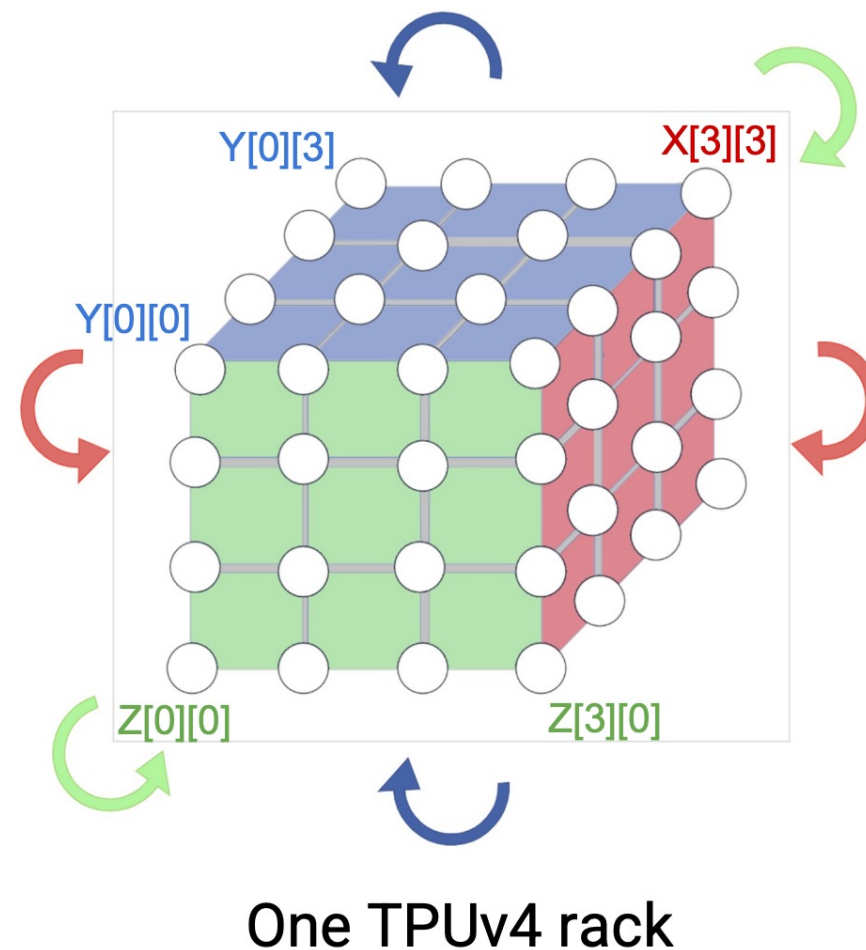
GPU clusters: Rail only (Meta)

Across each High Bandwidth Domain, only GPUs of the same index (same rail) are connect via rail interconnection



Google TPUv4 rack

Topology	3D torus
TPUs per Rack	64 (4x4x4)
Intra- rack connection	Inter-Chip Interconnection links 50 GB/s
Inter- rack connection	OCS links connected to surface nodes
Total TPUs per Pod	4096



Summary

- Data center
 - Huge server and network costs
 - Want to achieve agility and high resource utilization
- Server and network
 - Virtualization is a key
 - Data center network topology
- Emerging GPU clusters
 - New design for hardware and network to accommodate deep learning workloads

Credits

- Some slides adapted from course slides of COS461 in Princeton
- Some slides adapted from course slides of CSE599k in UW